

## **JNTU Kakinada**

### **MODEL QUESTION PAPER (R23 PATTERN)**

Model Question Paper 1 ? VLSI Design (Latest Pattern)

Subject Code: EC502 Semester: 5 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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#### **SECTION A (Unit 1: CMOS Fabrication and Layout)**

1. Compare and contrast the key concepts involved in CMOS Fabrication and Layout. [12 Marks]
2. Explain CMOS Fabrication and Layout with a neat diagram/sketch and a suitable example. [12 Marks]
3. Define CMOS Fabrication and Layout. Explain its significance in VLSI Design with a suitable example. [12 Marks]

#### **SECTION B (Unit 2: CMOS Inverter and Logic Gates)**

4. Explain CMOS Inverter and Logic Gates with a neat diagram/sketch and a suitable example. [12 Marks]
5. Solve a problem related to CMOS Inverter and Logic Gates. Show all the steps clearly. [12 Marks]
6. Solve a problem related to CMOS Inverter and Logic Gates. Show all the steps clearly. [12 Marks]

#### **SECTION C (Unit 3: Sequential Circuit Design)**

7. List the advantages and limitations of Sequential Circuit Design. Explain any two in detail. [12 Marks]
8. Derive the expression related to Sequential Circuit Design and explain each term. [12 Marks]
9. List the advantages and limitations of Sequential Circuit Design. Explain any two in detail. [12 Marks]

#### **SECTION D (Unit 4: Design for Testability)**

10. List the advantages and limitations of Design for Testability. Explain any two in detail. [12 Marks]
11. Solve a problem related to Design for Testability. Show all the steps clearly. [12 Marks]
12. List the advantages and limitations of Design for Testability. Explain any two in detail. [12 Marks]

#### **SECTION E (Unit 5: VLSI Design Flow)**

**13. Discuss the engineering applications of VLSI Design Flow. [8 Marks]**

14. Explain VLSI Design Flow with a neat diagram/sketch and a suitable example. [8 Marks]

15. Explain VLSI Design Flow with a neat diagram/sketch and a suitable example. [12 Marks]

\*\*\* End of Question Paper \*\*\*